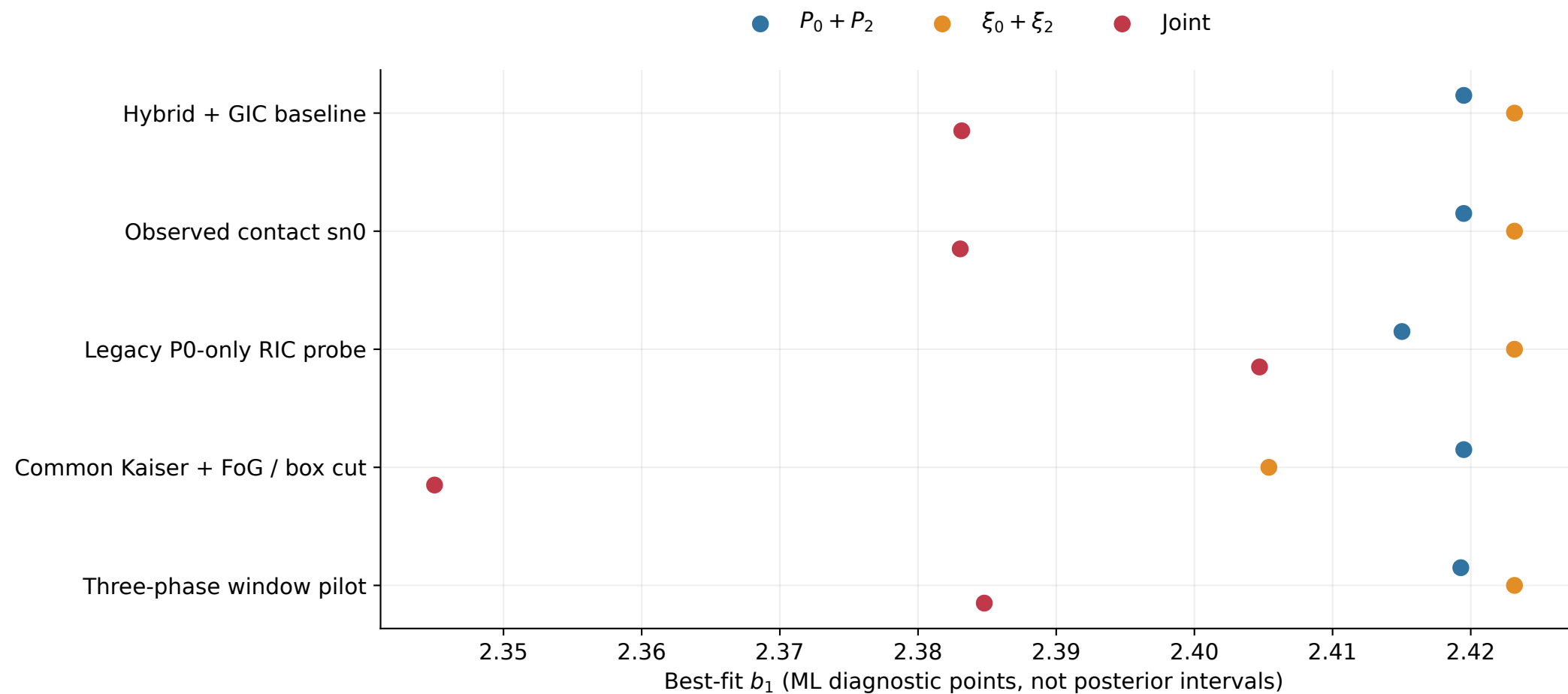


Which effects can move the joint bias?

EZmock1000 | 25-phase mean data | frozen 74 bins | full cross covariance | ML diagnostics

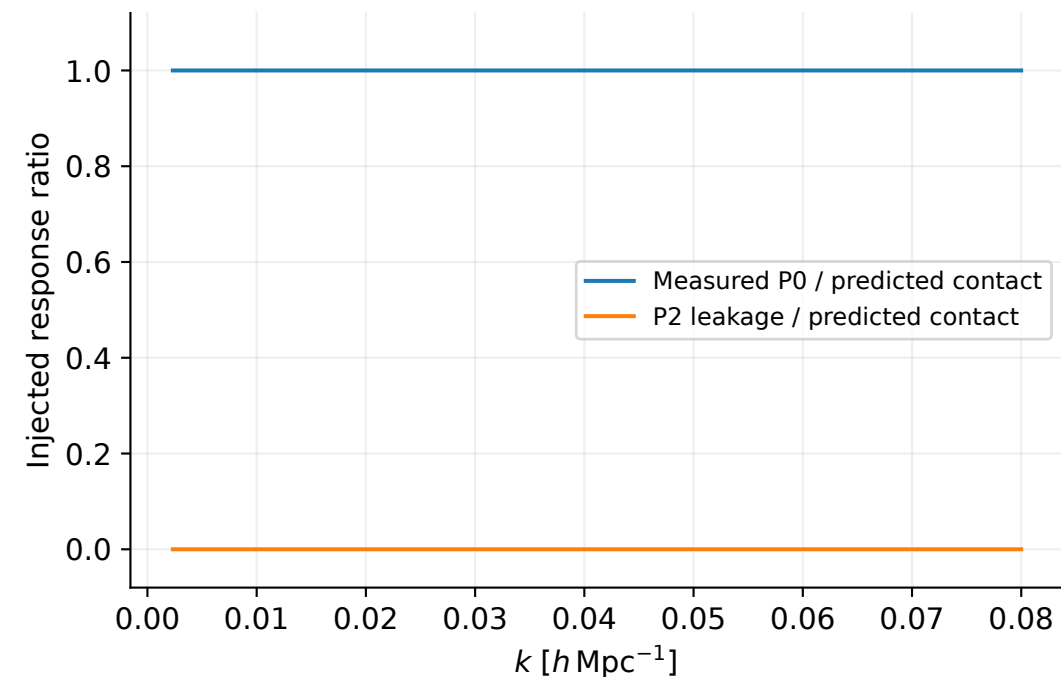
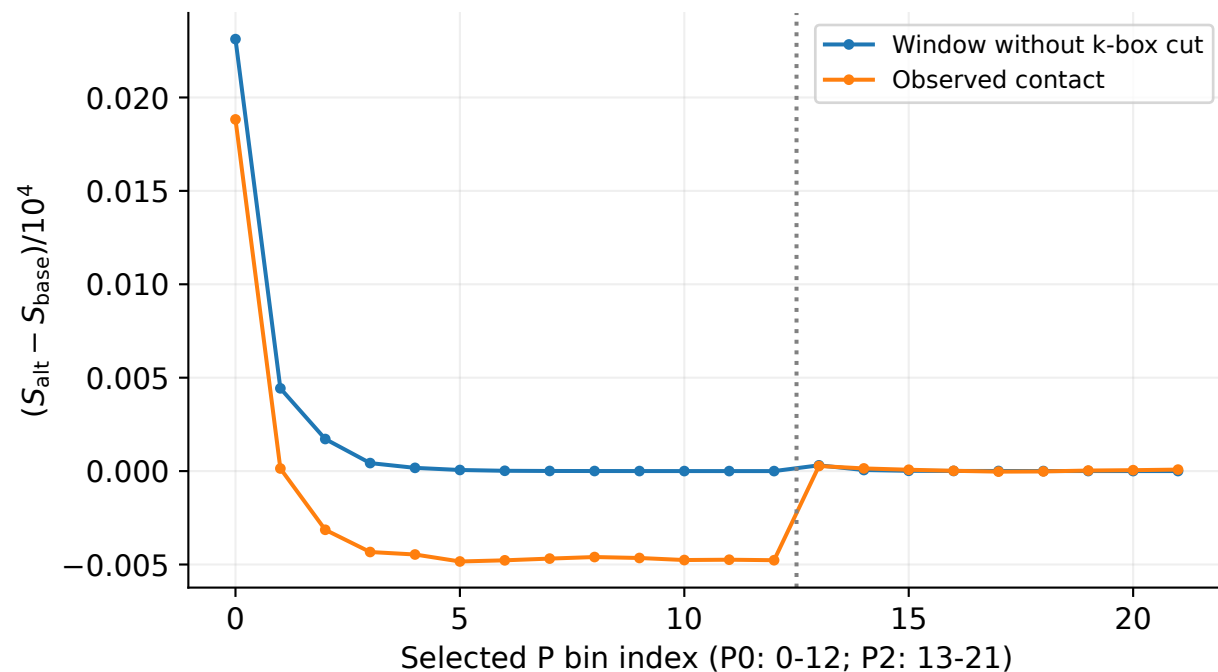


	Baseline	Contact	RIC probe	Common theory	Window pilot
Joint χ^2	4.5191	4.5182	4.6118	8.0104	4.4998
Joint sn_0	0.0824	0.0831	0.0358	0.1483	0.0800

RIC probe: a substantial bias shift with essentially unchanged fit quality (delta chi-square = +0.093). It is an incomplete P0-only approximation, so the formal hybrid + GIC result is retained.

Shot response: metadata and estimator injection both close

EZmock1000 | 25-phase mean data | frozen 74 bins | full cross covariance | ML diagnostics

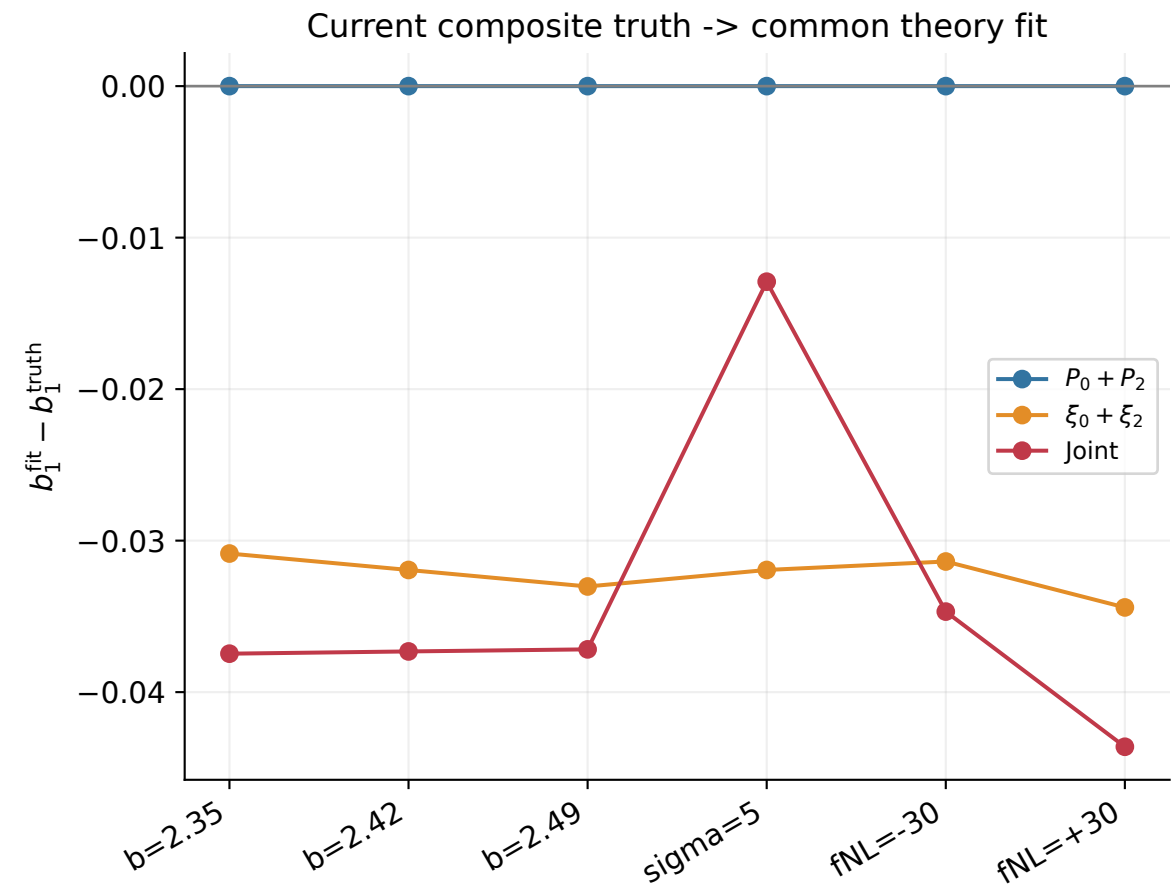
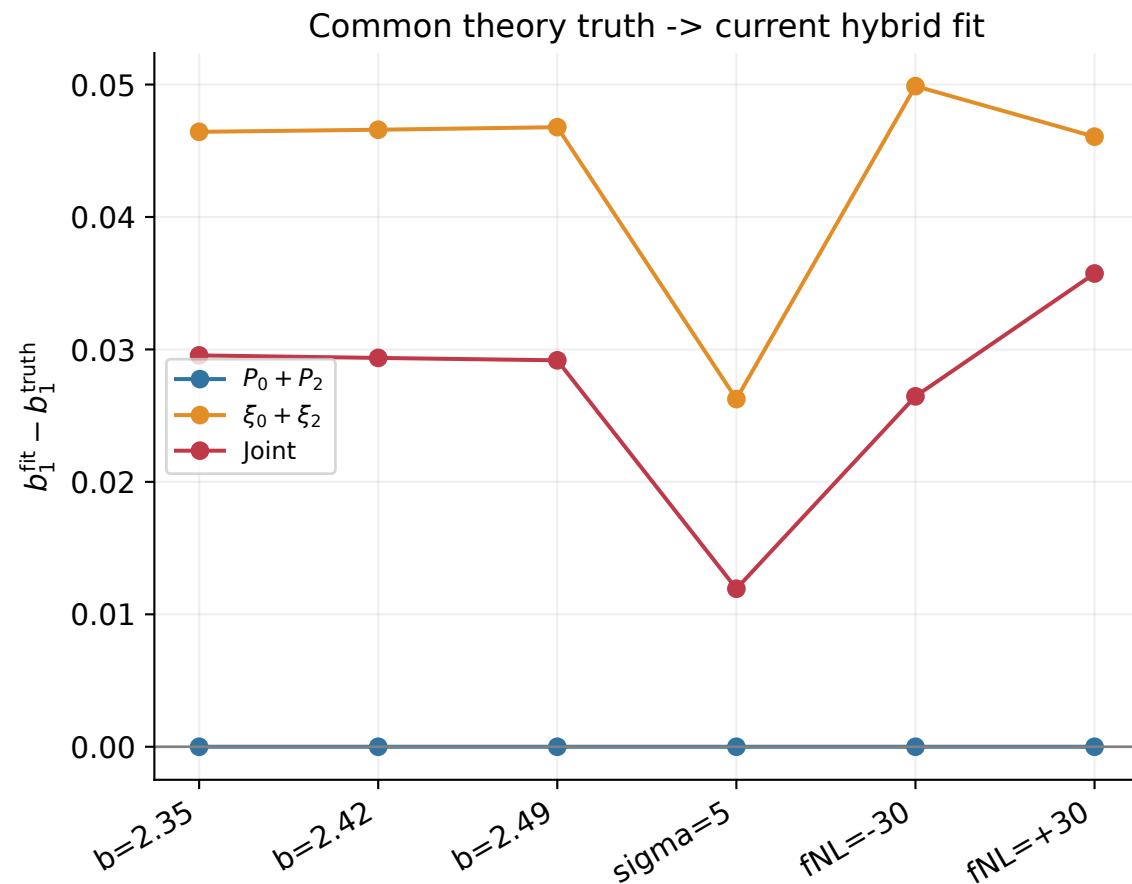


All 25 phase metadata: maximum shot-normalization relative error = $3.49\text{e-}13$.
Coincident half-weight pair test: P0 relative error = $1.95\text{e-}15$; P2 leakage = $1.35\text{e-}15$.
The painted field is unchanged; only the analytically subtracted self-pair term changes.
Replacing the fitted sn0 template shifts joint b1 by only -0.000112.

This validates the contact implementation. It does not prove that halo stochasticity is scale independent.

Cross-model injections: the direction of mismatch matters

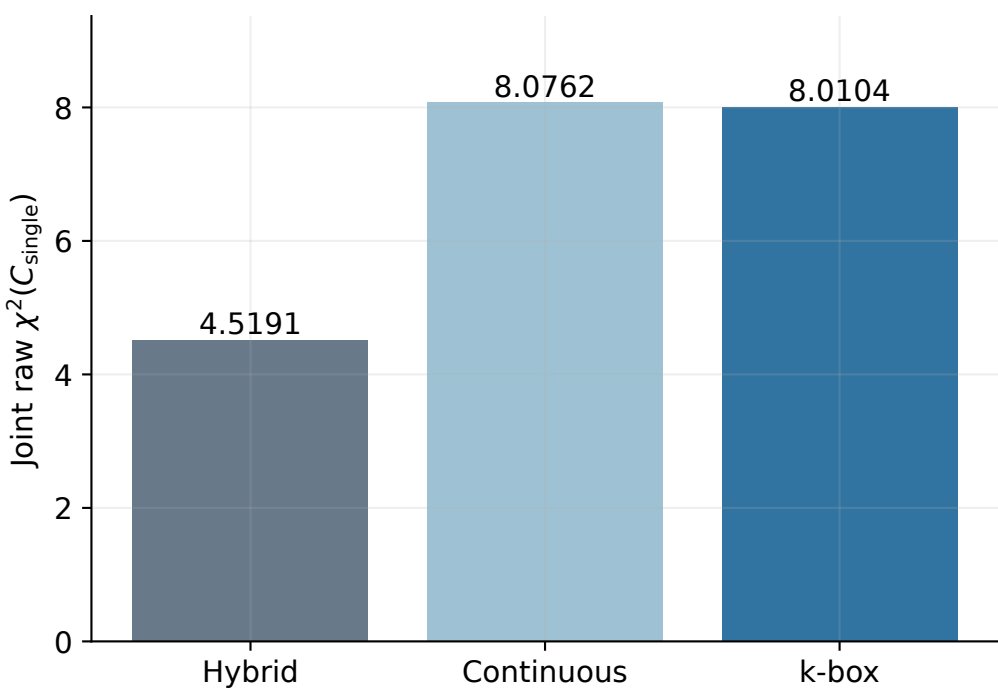
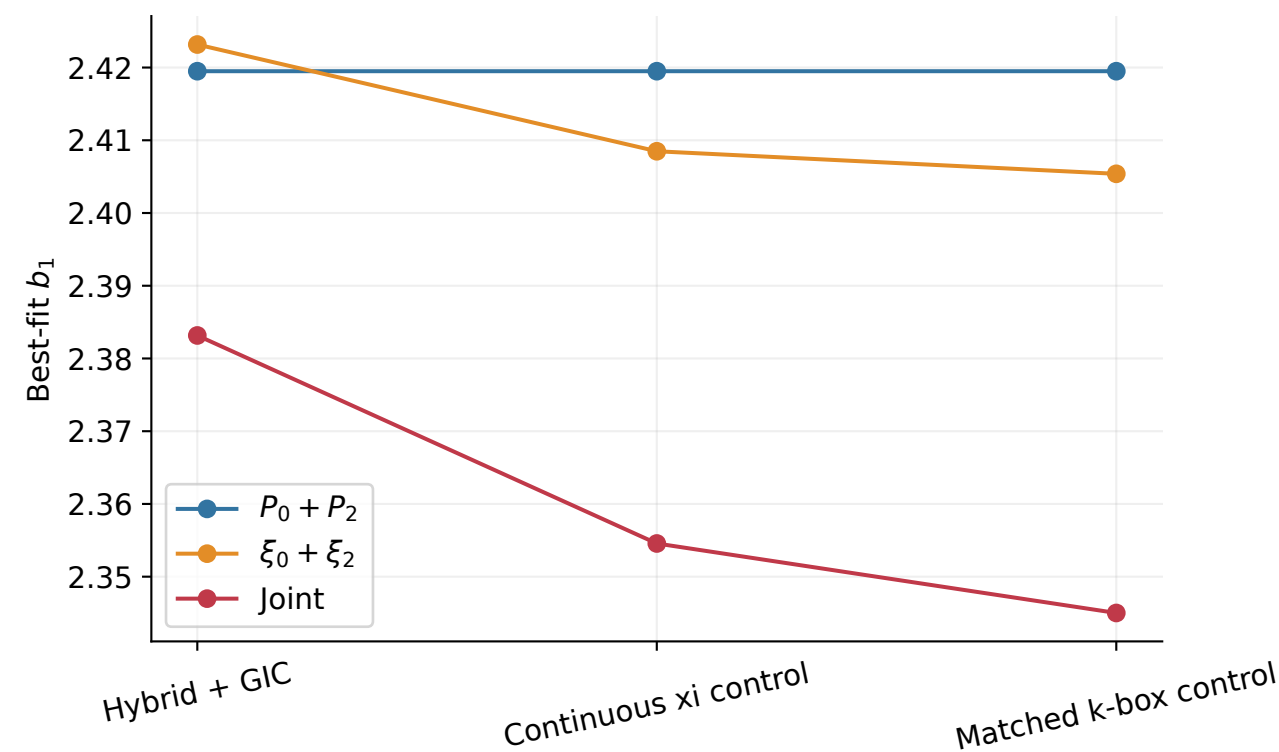
EZmock1000 | 25-phase mean data | frozen 74 bins | full cross covariance | ML diagnostics



Matched clustering control: common P_L , PNG response, Kaiser x Lorentzian² FoG, k-box support and xi GIC.
Forward injection: all six tested cases put joint b_1 between P and ξ ; the observed ordering is not reproduced.
Reverse injection can put joint below both. This is evidence of directional model sensitivity, not a preferred truth.
Self-recovery passes at all anchors; highest integration/interpolation discrepancy is $< 5e-9$ in $\chi^2(C_{\text{mean}})$.
Noiseless controls only. Common-theory ξ has a shared σ parameter; joint still has four parameters.

Common theory on the measured data does not resolve the offset

EZmock1000 | 25-phase mean data | frozen 74 bins | full cross covariance | ML diagnostics

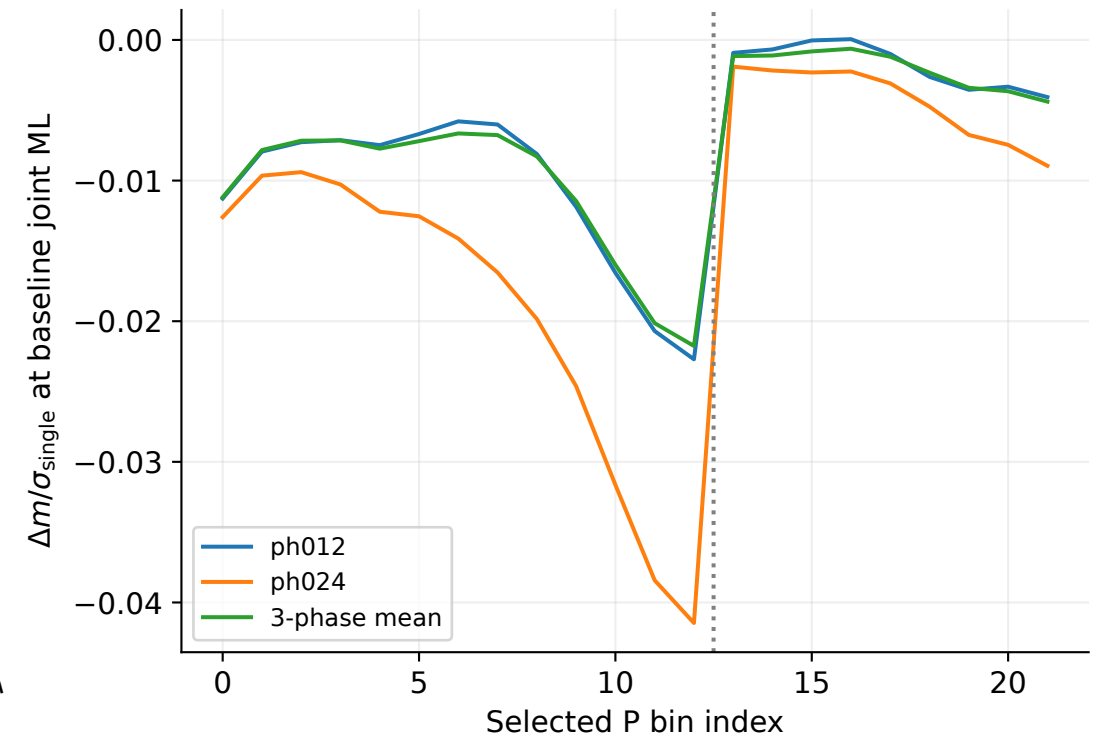
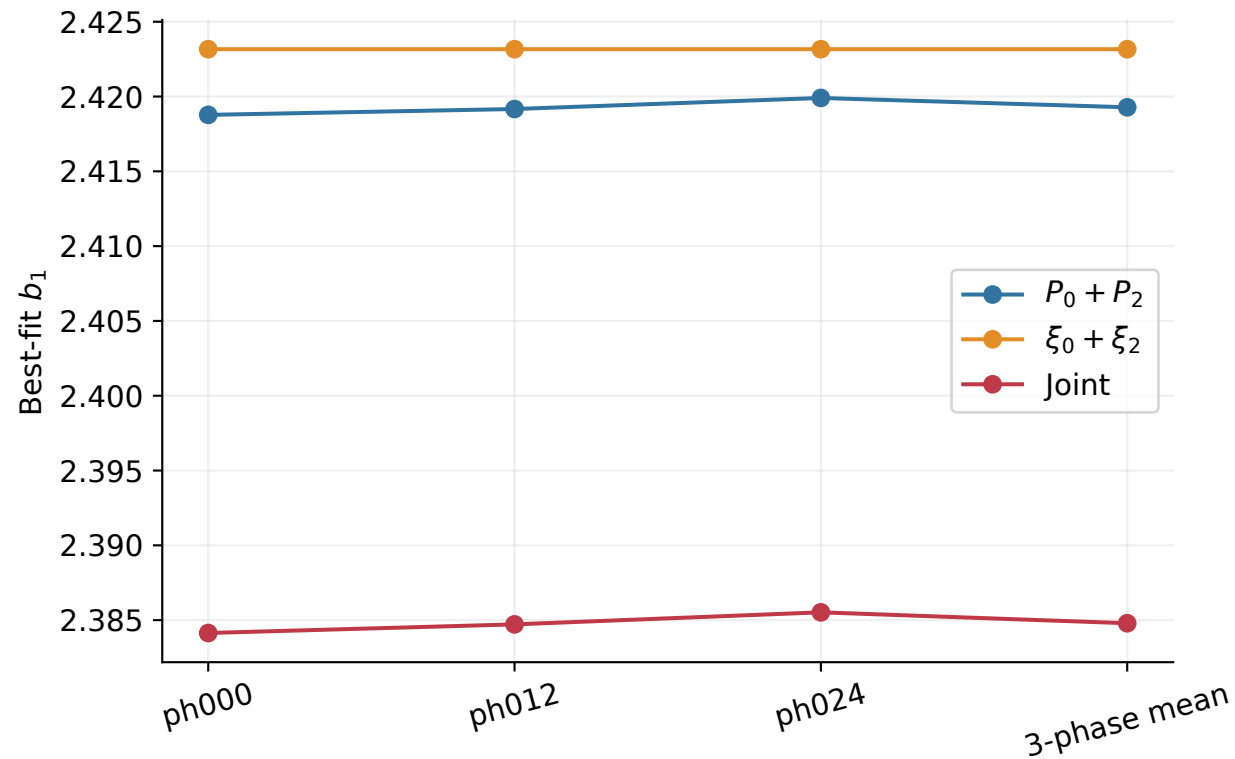


The primary common-theory control is the k-box branch; the continuous-xi branch leaves a support mismatch. Both controls move the real-data joint b_1 farther down. A common simple Kaiser model is not a successful repair. This test does not establish that the frozen hybrid is physically complete; it tests one specific mismatch.

The shared sigma is already present in P and joint; xi-only has three free parameters here versus two in hybrid. The P residual stochastic contact remains a separate nuisance and is not Hankel-transformed as clustering.

Phase-window pilot at the production estimator settings

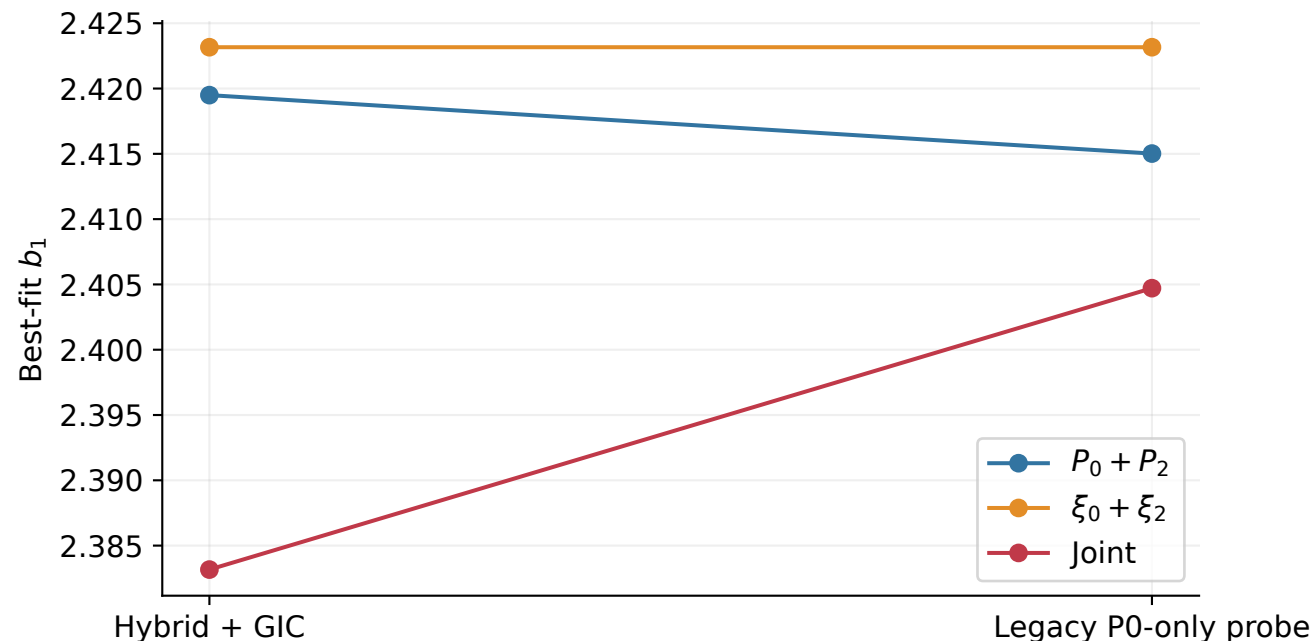
EZmock1000 | 25-phase mean data | frozen 74 bins | full cross covariance | ML diagnostics



P0/P2 data reproduce (max relative error 0.00e+00); the historical ph000 window does NOT reproduce exactly.
Independent current ph000 repeat agrees to relative 0.00e+00.
Window recomputation drift alone: delta $b_1 = +0.000976$; 3-phase mean versus current ph000: +0.000653.
Total 3-phase-pilot shift versus frozen historical window: +0.001629.
The current library was modified after the historical file; historical library sources were not archived here.
This is not the full 25-phase average. The historical implementation drift is not a phase effect.

RIC: a sensitive direction, with an incomplete existing operator

EZmock1000 | 25-phase mean data | frozen 74 bins | full cross covariance | ML diagnostics



What this probe actually does

Theory $\ell=0 \rightarrow$ observed P_0 only.
 P_2 , ξ_0 and ξ_2 receive no RIC term.
 No new fit parameter.

Joint b_1 : 2.3832 $\rightarrow$ 2.4047
 Joint raw chi-square: 4.5191 $\rightarrow$ 4.6118
 Fit quality is essentially unchanged.

Full field projection: $C_{\text{rad}} = (I - \Pi_r)C(I - \Pi_r)^T$

$$C_{\text{rad}} = C - \Pi_r C - C \Pi_r^T + \Pi_r C \Pi_r^T$$

The legacy approximation subtracts the radial auto contribution as an approximation to the net correction.
 The two cross terms are not evaluated; its accuracy for this geometry and joint vector is unvalidated.
 A complete radial projection includes the global mean constraint when their weights are consistent.
 Therefore, do not blindly add full RIC on top of the current scalar GIC prescription.
 Next priority: paired selection-function closure and a consistent P/ξ RIC response, with FKP held fixed.
 Reference: de Mattia & Ruhlmann-Kleider (2019), arXiv:1904.08851, equations (6), (20), section 2.4.